

## AutoDiag AI Test Diagnostic Document

Vehicle System: Engine

DTC: P0301

Description:

P0301 indicates that a misfire has been detected on cylinder 1.

Possible diagnostic areas:

- Ignition system
- Fuel delivery
- Injector operation
- Engine mechanical condition
- Air or vacuum leaks
- Wiring and electrical connections

Recommended diagnostic approach:

1. Confirm the DTC and check for additional codes.
2. Inspect ignition components associated with cylinder 1.
3. Check whether the misfire follows an ignition component when the applicable service procedure permits this test.
4. Check fuel injector operation and related wiring.
5. Check for intake or vacuum leaks.
6. Perform appropriate engine mechanical tests if needed.

Parts replacement:

Do not replace an ignition coil, spark plug, injector, or other component solely because P0301 is present. Confirm the failed component through appropriate testing.

Important:

P0301 is a diagnostic clue and does not by itself prove that a specific component has failed.

Customer explanation:

The vehicle computer has detected a misfire affecting cylinder 1. We need to perform checks to determine the underlying cause before replacing parts.